



NORTH SYDNEY BOYS HIGH SCHOOL

MATHEMATICS

2010 HSC Course Assessment Task 2

General instructions

- Working time – 65 minutes.
- Write in the booklet provided.
Each question is to commence on a new page.
- Write using blue or black pen.
- Board approved calculators may be used.
- All necessary working should be shown in every question (Insufficient/illegible working may cause a deduction of marks).
- Attempt **all** questions.

Class teacher (please ✓)

- ☐ 12M21 – Mr Berry
- ☐ 12M22 – Mr Fletcher
- ☐ 12M31 – Mr Rezcallah
- ☐ 12M32 – Mr Lowe
- ☐ 12M33 – Mr Ireland
- ☐ 12M41 – Mr Barrett
- ☐ 12M42 – Mr Trenwith
- ☐ 12M43 – Mr Weiss

STUDENT NUMBER:

Marker's use only.

QUESTION	1	2	3	4	5	6	7	Total	%
MARKS	$\overline{10}$	$\overline{6}$	$\overline{9}$	$\overline{10}$	$\overline{8}$	$\overline{12}$	$\overline{12}$	$\overline{67}$	

Question 1 (10 Marks)	Commence a NEW page.	Marks
(a) Find:		
i. $\int (x^3 - 3x^2 + 4) dx.$		2
ii. $\int (x + 2)(2x - 5) dx.$		2
iii. $\int \frac{3x^5 + 2x^3 - 1}{x^2} dx.$		2
(b) Evaluate:		
i. $\int_0^1 x\sqrt{x} dx.$		2
ii. $\int_2^3 (2x - 5)^3 dx.$		2

Question 2 (6 Marks)	Commence a NEW page.	Marks
(a) The gradient function of a curve is given by $\frac{dy}{dx} = 2x + 5$. If $y = -1$ when $x = 2$, find y as a function of x .		3
(b) Find the values of k for which		3
$\int_1^k (x + 1) dx = 6$		

Question 3 (9 Marks)	Commence a NEW page.	Marks
(a) A parabola has its vertex at the point $(3, 1)$ and its directrix has equation $y = -1$.		
i. What is its focal length?		1
ii. State the coordinates of the focus.		1
iii. Find the equation of the parabola.		1
iv. What is the equation of its axis of symmetry?		1
(b) A parabola has equation $x^2 + 2x + 8y + 25 = 0$. Write down:		
i. the coordinates of its vertex.		2
ii. the coordinates of its focus.		2
iii. the equation of its directrix.		1

Question 4 (10 Marks)

Commence a NEW page.

Marks

- (a) If
- α
- and
- β
- are the roots of
- $2x^2 + 3x - 4 = 0$
- , find the value of

i. $\alpha + \beta$.

1

ii. $\alpha\beta$

1

iii. $(\alpha - 3)(\beta - 3)$.

2

- (b) For what values of
- p
- will the equation

2

$$2px^2 - (p + 2)x + (p - 4) = 0$$

have roots which are reciprocals of each other?

- (c) Solve for
- x
- :

4

$$(x^2 - 2x)^2 - (x^2 - 2x) - 6 = 0$$

Question 5 (8 Marks)

Commence a NEW page.

Marks

- (a) Find the values of
- m
- for which

2

$$5x^2 - 4x + m = 0$$

has real roots.

- (b) Find the values of
- k
- for which

3

$$x^2 - (k + 2)x + (3k + 6)$$

is positive definite.

- (c) Find the values of
- A
- ,
- B
- and
- C
- for which

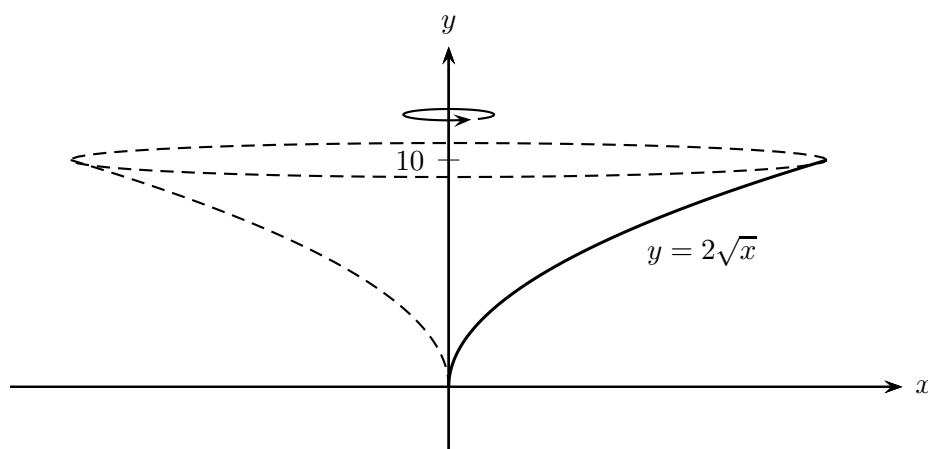
3

$$9x^2 + 2x - 5 \equiv Ax(x + 1) + B(x + 1) + C$$

Question 6 (12 Marks)		Commence a NEW page.	Marks
(a)	Calculate the area of the region between the curve $y = x^2 - 2x$, the x axis and the lines $x = 1$ and $x = 3$.		4
(b)	i. Show that the points of intersection of the line $y = 2x + 5$ and the parabola $y = x^2 - 2x$ are $(5, 15)$ and $(-1, 3)$.		2
	ii. Calculate the area between the line $y = 2x + 5$ and the parabola $y = x^2 - 2x$.		3
(c)	Use Simpson's Rule with five function values to find an approximation to		3

$$\int_1^3 3^{x-1} dx$$

Question 7 (12 Marks)		Commence a NEW page.	Marks
(a)	i. Find the locus of the point $P(x, y)$ which moves such that its distance from the point $A(1, -2)$ is always twice the distance from the point $B(7, -8)$.		3
	ii. Describe this locus geometrically, clearly stating its features.		2
(b)	A glass is obtained by rotating part of the parabola $y = 2\sqrt{x}$ about the y axis as shown.		3



The glass is 10 cm deep.

Find the volume of liquid that the glass will hold.

(c)	i. Draw a neat sketch of the parabola $x^2 = -4ay$.	1
	ii. The area between the parabola and its latus rectum is rotated about the x axis. Calculate the volume of the solid which is generated.	3

End of paper.

QUESTION 1

(a) (i) $\int (x^3 - 3x^2 + 4) dx = \frac{x^4}{4} - x^3 + 4x + c$

2

no c - 1

(ii) $\int (x+2)(2x-r) dx = \int (2x^2 \cdot x - 10) dx$
 $= \frac{2}{3}x^3 - \frac{x^2}{2} - 10x + c$

1

1

(iii) $\int \frac{3x^5 + 2x^3 - 1}{x^2} dx = \int (3x^3 + 2x - x^{-2}) dx$
 $= \frac{3}{4}x^4 + x^2 + x^{-1} + c$

1

1

(b) (i) $\int_0^1 x\sqrt{x} dx = \int_0^1 x^{3/2} dx$
 $= \left[\frac{2}{5} x^{5/2} \right]_0^1$
 $= \frac{2}{5} - 0 = \frac{2}{5}$

1

1

(ii) $\int_2^3 (2x-r)^3 dx = \left[\frac{(2x-r)^4}{8} \right]_2^3$
 $= \left(\frac{1}{8} - \frac{(-1)^4}{8} \right) = \frac{1}{8} - \frac{1}{8} = 0$

1

1

QUESTION 2

(a) $\frac{dy}{dx} = 2x + r$
 $y = x^2 + rx + c$

1

When $x=2, y=-1 \Rightarrow -1 = 2^2 + r \cdot 2 + c$
 $c = -1r$

1

1

$\therefore y = x^2 + rx - 1r$

(b) $\int_1^k (x+1) dx = \left[\frac{x^2}{2} + x \right]_1^k$
 $= \left(\frac{k^2}{2} + k \right) - \left(\frac{1}{2} + 1 \right)$
 $= \frac{k^2}{2} + k - \frac{3}{2} = 6$

1

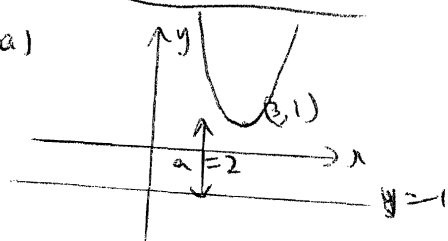
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i.e. $k^2 + 2k - 3 = 12$
 $k^2 + 2k - 15 = 0$
 $(k+5)(k-3) = 0$
 $k = -5 \text{ or } k = 3$

1

QUESTION 3

(a)



(i) $a = 1 - (-1) = 2$ |

(ii) focus is at (3, 3) |

(iii) equation of parabola is
 $(x-3)^2 = 8(y-1)$ |

(iv) axis of symmetry
 has equation $x = 3$ |

(b) (i) $x^2 + 2x + py + 25 = 0$

$x^2 + 2x + 1 = -py - 24$

$(x+1)^2 = -p(y+3)$ |

∴ vertex is at $(-1, -3)$ |

(ii) $a = 2$ |

∴ focus is at $(-1, -5)$ |

(iii) directrix has equation $y = -1$ |

QUESTION 4

(a) (i) $\alpha + \beta = -\frac{b}{a} = -\frac{3}{2}$ |

(ii) $\alpha\beta = \frac{c}{a} = -\frac{4}{2} = -2$ |

(iii) $(\alpha-3)(\beta-3) = \alpha\beta - 3\alpha - 3\beta + 9$
 $= \alpha\beta - 3(\alpha + \beta) + 9$
 $= -2 - 3 \times -\frac{3}{2} + 9$
 $= \frac{23}{2}$ |

(b) $\alpha\beta = 1$
 i.e. $\frac{c}{a} = 1$
 i.e. $\frac{p-4}{2p} = 1$ |
 $p-4 = 2p$
 $p = -4$ |

(c) Let $u = x^2 - 2x$
 then equation becomes $u^2 - u - 6 = 0$

$(u-3)(u+2) = 0$

$u = 3$ or $u = -2$ |

i.e. $x^2 - 2x = 3$ or $x^2 - 2x = -2$

$x^2 - 2x - 3 = 0$ or $x^2 - 2x + 2 = 0$

$(x-3)(x+1) = 0$ |
 $x = 3, x = -1$ 2 solutions

∴ valid solutions are $x = 3$ or $x = -1$

QUESTION 5

(a) real roots $\Rightarrow \Delta \geq 0$

$$\text{i.e. } (-4)^2 - 4 \times 5 \times m \geq 0 \quad |$$

$$16 - 20m \geq 0$$

$$20m \leq 16$$

$$m \leq \frac{4}{5} \quad |$$

(b) positive definite $\Rightarrow a > 0, \Delta < 0$ |

$$a = 1 > 0$$

$$\Delta = (k+2)^2 - 4(3k+6)$$

$$= k^2 + 4k + 4 - 12k - 24$$

$$= k^2 - 8k - 20$$

$$= (k-10)(k+2) \quad |$$

$$\text{we want } (k-10)(k+2) < 0$$

$$\text{i.e. } -2 < k < 10 \quad |$$

(c) $9x^2 + 2x - 7 \equiv A x(x+1) + B(x+1) + C$

$$\text{sub } x = -1 \Rightarrow C = 9 - 2 - 7 = 2 \quad |$$

$$\text{sub } x = 0 \Rightarrow B + C = -7 \Rightarrow B = -9 \quad |$$

$$\text{equate coefficients of } x^2 \Rightarrow A = 9 \quad |$$

OR

$$A x(x+1) + B(x+1) + C = Ax^2 + Ax + Bx + B + C$$

$$= Ax^2 + (A+B)x + (B+C)$$

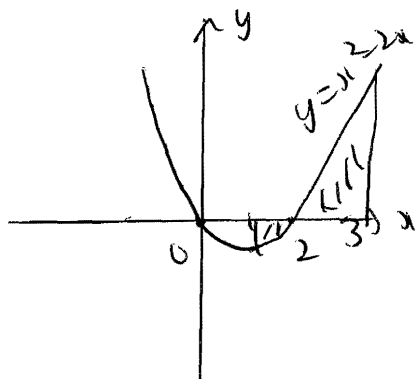
$$\text{equate coefficients of } x^2 \Rightarrow A = 9 \quad |$$

$$\text{equate coefficients of } x \Rightarrow A+B = 2 \Rightarrow B = -7 \quad |$$

$$\text{equate constants} \Rightarrow B+C = -7 \Rightarrow C = 2 \quad |$$

QUESTION 6

(a)



$$\begin{aligned} \text{Required area} &= \int_2^3 (x^2 - 2x) dx + \left| \int_1^2 (x^2 - 2x) dx \right| \\ &= \left[\frac{x^3}{3} - x^2 \right]_2^3 + \left| \left[\frac{x^3}{3} - x^2 \right]_1^2 \right| \\ &= \left(9 - 4 - \frac{8}{3} + 4 \right) + \left| \left(\frac{8}{3} - 4 - \frac{1}{3} + 1 \right) \right| \\ &= \frac{4}{3} + \left| -\frac{2}{3} \right| \\ &= \frac{4}{3} + \frac{2}{3} = 2 \text{ units}^2 \end{aligned}$$

(b)

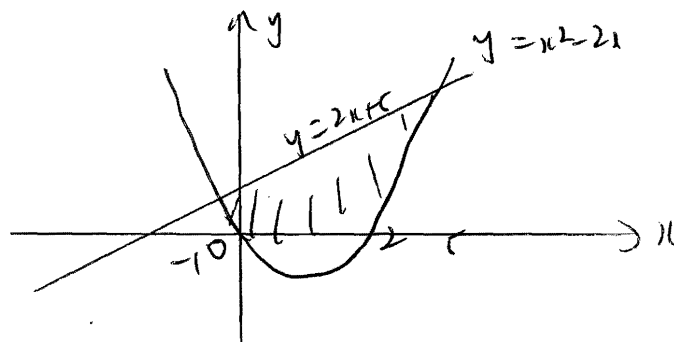
$$\begin{aligned} (i) \quad x^2 - 2x &= 2x + r \\ x^2 - 4x - r &= 0 \\ (x - r)(x + 1) &= 0 \\ x = r \text{ or } x = -1 \end{aligned}$$

$$\text{When } x = r, y = 2x + r = 1r$$

$$\text{When } x = -1, y = 2x - 1 + r = 3$$

i.e. points of intersection are $(r, 1r)$ and $(-1, 3)$

(ii)



$$\begin{aligned} \text{required area} &= \int_{-1}^r (2x + r - (x^2 - 2x)) dx \\ &= \int_{-1}^r (4x + r - x^2) dx \\ &= \left[2x^2 + rx - \frac{x^3}{3} \right]_{-1}^r \\ &= \left(5r + 2r - \frac{12r}{3} \right) - \left(2 - r + \frac{1}{3} \right) \\ &= \frac{10r}{3} + \frac{8}{3} = 36 \text{ units}^2 \end{aligned}$$

(c)

$$\begin{aligned} \int_1^3 3^{x-1} dx &= \frac{3-1}{12} \left(3^0 + 4 \times 3^{\frac{1}{2}} + 2 \times 3^1 + 4 \times 3^{\frac{3}{2}} + 3^2 \right) \\ &= \frac{1}{6} (1 + 4\sqrt{3} + 6 + 12\sqrt{3} + 9) \\ &= \frac{16 + 16\sqrt{3}}{6} = \frac{8 + 8\sqrt{3}}{3} \\ &= \frac{8(1 + \sqrt{3})}{3} \end{aligned}$$

wrong rule - no marks

QUESTION 7

(a) (i) $PA^2 = (x-1)^2 + (y+2)^2$
 $PB^2 = (x-7)^2 + (y+8)^2$

$PA = 2PB$

$PA^2 = 4PB^2$

i.e. $x^2 - 2x + 1 + y^2 + 4y + 4 = 4x^2 - 56x + 196 + 4y^2 + 64y + 256$

$3x^2 - 14x + 3y^2 + 60y + 447 = 0$

$x^2 - 14x + y^2 + 20y + 149 = 0$

$(x-7)^2 + (y+10)^2 = -149 + 49 + 100$
 $= 32$

(ii) Curve is a circle centre (7, -10) radius $4\sqrt{2}$
 — 1 French marks

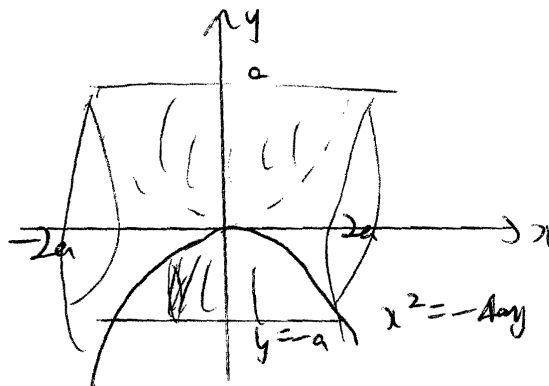
(b) (i) $V = \pi \int_0^{10} x^2 dy$ $y = 2\sqrt{x} \rightarrow y^2 = 4x$
 $x = \frac{y^2}{4}$
 $x^2 = \frac{y^4}{16}$

$= \pi \int_0^{10} \frac{y^4}{16} dy$

$= \left[\frac{\pi y^5}{80} \right]_0^{10}$

$= \frac{\pi}{80} \times 10^5 = \frac{2500\pi}{2} \text{ units}^3$

(c) (i)



(ii) required volume = volume of cylinder $- \pi \int_{-2a}^{2a} y^2 dx$

$= \pi \times a^2 \times 4a - 2\pi \int_0^{2a} \frac{x^4}{16a^2} dx$

$= 4\pi a^3 - 2\pi \left[\frac{x^5}{80a^2} \right]_0^{2a}$

$= 4\pi a^3 - \frac{4\pi a^3}{5} = \frac{16\pi a^3}{5} \text{ units}^3$